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18 June 1970
SUPERSEDING
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MILITARY SPECIFICATION

PULSE FORMING NETWORK POWER SUPPLY: 10540707

1. SCOPE

1.1 This specification covers a solid state power supply for charging energy storage capacitors to a preset d.c. voltage called the shutoff voltage. This power supply is a modular component of the Range Finder Laser, AN/GVS-3, (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical, for Coating Printed Circuit Assemblies

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10540501	Range Finder, Laser, AN/GVS-3

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F10540705	Right Housing
D10540707	PFN Power Supply Assembly
C10540748	Connector
C10540749	Connector
C10541003	Connector

PACKAGING DATA SHEETS

10540707	Packaging of Pulse Forming Network Power Supply

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The pulse forming network power supply hereinafter referred to as power supply shall be manufactured in accordance with Drawing D10540707 and drawings pertaining thereto.

3.3 General specification.- The power supply shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Electrical and electronic assemblies
(d) Soldering electrical connections
(e) Part identification and marking
(f) Workmanship

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926, the power supply shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.12.

3.3.1.1 Etched circuit board.- The conductor pattern on the etched board shall be free from blisters, wrinkles, cracks, holes, dents, scratches and any other defects that will affect serviceability. The pattern shall be free of loosely adhering feather edges of metal. In no case shall a finished line width be more than 0.010 inch narrower than that specified on the printed circuit master. In no case shall any conductor line width be less than 0.020 inch.

3.3.1.2 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with at least one point of the

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component in contact with the board. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface.

3.3.1.3 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that any reference symbols on the board are not obscured, and that the proper identification marking of each part is visible.

3.3.1.4 Transistors.- Transistors are to be mounted perpendicular to the circuit board with spacers between the transistors and the circuit board.

3.3.1.5 Bend radius.- The minimum bend radii for all formed leads shall be one lead diameter. The maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.3.1.6 Component leads.- Leads of transistors mounted with a spacer directly into a printed circuit board, shall have 1/16 inch of lead from the transistor base to any soldered point. Component leads shall be preformed to suit mounting holes, there shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than 3/64 inch including solder beyond the side of the circuit board opposite the component.

3.3.1.7 Plated through connections.- All plated through connections shall be filled with solder in order to eliminate dependence on plating itself for circuit continuity.

3.3.1.8 Heat sensitive components.- The use of heat shunts or cooling devices shall be mandatory when hand soldering components that are subject to damage by heat.

3.3.1.9 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed, the cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 pounds per square inch. The temperature of the cleaning solution shall not exceed 125°F. After cleaning there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.3.1.10 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than room temperature drying method is used the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psig. The air or nitrogen shall be filtered to meet the following requirements:

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(a) Permissible dust particle size not to exceed 6 microns
(b) Permissible moisture content limited to 2.12 grains per cubic foot (dew point 32°F maximum)
(c) Dust count shall not exceed 1000 particles per cubic foot

3.3.1.11 Cementing and coating.

3.3.1.11.1 Masking.- Prior to coating the printed circuit board assembly with insulating compound MIL-I-45608, the high voltage output terminals, the identified test points shall be masked to protect them from being coated. Subsequent to coating and when masking is removed the above parts shall be clean and free of any coating.

3.3.1.11.2 Coating, insulating compound.- The printed circuit board assembly, with electrical components installed, shall be coated with an insulating compound. The insulating compound shall be Insulating Compound, Electrical, Type PUR, Polyurethane Resin conforming to MIL-I-46058. Neither heat sink (top, bottom and sides) nor the electronic components mounted thereon, shall be coated with resin. The exterior wire harness and the lead wires between the heat sink and the board and potentiometer R16 and the board shall not be coated with resin.

3.3.1.11.3 Cured compound.- The cured compound shall represent a smooth continuous surface and shall pass from the mounted components to the circuit board with no break at their junction. Except for areas masked, all surfaces shall be covered with a coating of 0.001 to 0.006 inch thickness. The cured compound shall meet the following requirements:

(a) Compound shall be free of visible dust particles or other foreign matter.
(b) It shall be smooth and uniform and have no globules or bare spots.
(c) It shall be substantially free of bubbles.
(d) There shall be no blisters or cracks.
(e) It shall be transparent and shall not affect marking legibility.
(f) It shall be capable of being removed in local areas for replacement of components by a soldering iron or similar method.
(g) Applied to a test board of 1/16 inch thickness, the compound shall not rupture or separate from the board, when the board is bent into an arc of twelve inches radius.

3.3.1.12 Wire termination.- When soldering wires to terminals or solder points a small gap of 1/32 inch minimum to 1/16 inch maximum shall be left between the wire insulation and the solder build-up. Burned, frayed, or otherwise damaged wire insulation shall not be acceptable. Stripping of insulation shall be accomplished without disarranging, cutting, or otherwise damaging any strand of the conductor. Heat shall not be used to remove outer jacket from shielded wire. Only mechanical methods of removal are permitted.

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EXCEPTION: The hot wire stripping method may be used where the wire or cable primary insulation and jacket are made of Teflon (TFE) or equal.

3.4 Vibration.- The power supply shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied continuously from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequency. Subsequent to vibration, the pulse forming network power supply shall show no evidence of physical damage, and shall meet the requirements of 3.7, 3.8.6, 3.9 and 3.10.

3.5 Shock.- The power supply shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular directions. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the pulse forming network power supply shall show no evidence of physical damage and shall meet the requirements of 3.7, 3.8.6, 3.9 and 3.10.

3.6 Storage temperatures.- The power supply shall withstand thermal stabilization at temperatures of -80°Fahrenheit (F) and +160°F. Subsequent to exposure to the temperatures specified herein, the pulse forming network power supply shall show no evidence of physical damage, and shall meet the requirements of 3.7, 3.8.6, 3.9 and 3.10.

3.7 Operational temperatures.- The power supply shall meet the requirements of 3.8.1 through 3.8.5 when operation is started from a thermally stabilized condition at temperatures of -65°F, +155°F and at a room temperature of between +70° and +80°F.

3.8 Performance.

3.8.1 Operational cycle.- Each requirement of 3.8.2 to 3.8.6 shall be met with the power supply operated through 12 charging cycles performed in rapid succession charging a 75mf ±5% capacitor. With potentiometer R16 remaining set in a given position, the variation between the minimum and maximum values of all shutoff voltages obtained from all combinations of input voltage and operating temperature shall not exceed 10 volts.

3.8.2 Shutoff voltage.

(a) With input voltages of 21 ± 0.1, 26.5 ± 0.1 and 30 ± 0.1 VDC, the pre-set shutoff voltage shall be greater than 900 VDC but less than 1100 VDC when potentiometer R16 is at its full counterclockwise position.

(b) With input voltage of 21 ± 0.1, 26.5 ± 0.1, and 30 ± 0.1 VDC, the pre-set shutoff voltage shall be greater than 1900 VDC, but less than 2050 VDC when potentiometer R16 is at its full clockwise position.

3.8.3 Capacitor charge time.- The time required for the power supply to charge the energy storage capacitor to the pre-set shutoff voltage shall not exceed 10 seconds when operated under the conditions of 3.8.2.

3.8.4 Input current.

(a) The input current of the power supply in a quiescent state shall not exceed 25 milliamps.

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(b) The input current of the power supply while charging the capacitor shall not exceed 4.0 amperes when checked at each of the requirements of 3.8.2.

3.8.5 Pulse output.- A positive pulse of at least 3 volt amplitude and 1 ± .5 milliseconds duration, measured at the half amplitude points, shall appear at pin L of P1 when 6 volts is applied to pin H. The power supply shall be capable of generating only one pulse at the completion of a charging cycle.

3.8.6 Program shutoff voltage.- Subsequent to meeting the requirements of 3.8.1 through 3.8.5, the pulse forming network power supply with an input voltage of 26.5 volts shall be adjusted, via potentiometer knob R16, to a shutoff voltage, Vo, of 1500 ± 5 volts when a resistor of 13,600 ± 250 ohms is connected across contacts J and K of P-1. With the shutoff voltage set to a Vo as indicated above and an input voltage of 26.5 volts, the following changes in shutoff voltage shall occur when the 13.6K resistor is changed to the values indicated below:

(a) Replacement of the 13.6K ohm resistor with a 7100 ± 250 ohm resistor between contacts J and K of P-1, shall cause a change in shutoff voltage of (Vo + 250) ± 5 volts .

(b) Replacement of the 13.6K ohm resistor with an 18,100 ± 300 ohm resistor between contacts J and K of P-1, shall cause a change in shutoff voltage of (Vo - 135) ± 5 volts .

3.8.7 Final voltage setting.- Subsequent to meeting all the preceeding requirements, the R16 potentiometer shall be adjusted to produce a shutoff voltage of between 1550 and 1600 volts, when the power supply is energized with 26.5 ± 0.1 volts at a temperature of between +70°F and +80°F with a 13.6K resistor connected between contacts J and K of P1.

3.9 Electrical continuity.- The electrical wiring of the pulse forming network power supply shall conform to the electrical shcematic dwg. D10544371.

3.10 Interface requirements.- The power supply shall interface with:

(a) Mounting flange and holes of Housing, 10540705
(b) Connector, 10540749
(c) Connector, 10541003

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) power supplies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Electronic workmanship	3.3.1	Visual
102. Etched circuit board	3.3.1.1	SME, Visual
103. Spacing of parts	3.3.1.2	SME, Visual
104. Symbols and markings	3.3.1.3	Visual
105. Transistors	3.3.1.4	Visual
106. Bend radius	3.3.1.5	SME, Visual
107. Component leads	3.3.1.6	SME, Visual

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108.	Plated through connections	3.3.1.7	SME, Visual
109.	Heat sensitive components	3.3.1.8	Visual
110.	Masking	3.3.1.11.1	Visual
111.	Cured compound	3.3.1.11.3	SME, Visual
112.	Wire termination	3.3.1.12	SME, Visual

MINOR: NONE

NOTE: The above characteristics shall be inspected at in-process intermediate inspection stations.

TABLE II - CLASSIFICATION OF DEFECTS
(100% Inspection)

		REQUIREMENT	TEST PROCEDURE
113.	Operational cycle	3.8.1	4.6.2
114.	Shutoff voltage	3.8.2	4.6.2.2
115.	Capacitor charge time	3.8.3	4.6.2.1
116.	Input current	3.8.4	4.6.2.3
117.	Pulse output	3.8.5	4.6.2.4
118.	Program shutoff voltage	3.8.6	4.6.2.5
119.	Final voltage setting	3.8.7	4.6.2.6
120.	Electrical continuity	3.9	4.6.3
121.	Interface requirements	3.10	4.6.4

NOTE: The tests in Table II shall be conducted at a temperature between +70° and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.1.9
Drying	3.3.1.10
Coating, insulating compound	3.3.1.11.2

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4.4 Special sampling.

4.4.1 General.- Three power supplies shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable Dwgs & Specs - Visual
302.	Fabrication	3.2	Applicable Dwgs & Specs - Visual
303.	General specifications	3.3	MIL-F-13926 - Visual

NOTE: The tests in Table III shall be conducted at a temperature between +70° and +80°F.

4.4.2 Environmental.- Three power supplies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall be subjected to the tests in Table IV and shall then meet the requirements and tests of Table II.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperatures (-80°F & +160°F)	3.6	4.6.1.1
305.	Operational temperatures (-65°F and +155°F)	3.7	4.6.1.2
306.	Shock (between +70°F and +80°F)	3.5	4.6.1.3
307.	Vibration (between +70°F & +80°F)	3.4	4.6.1.4

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

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4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests specified in 4.6:

	NOMENCLATURE	DESCRIPTION
1.	DC Power Supply, Regulated	Range of 0-30 volts, current rating 5 amps,
2.	DC Power Supply	6 volts ± 5%, 1 amp
3.	High voltage isolating diode	TRW semiconductor #PS-1851 or equivalent
4.	Resistance (one each)	13.6K ± 1%, 1/4 watt; 10K ±5%, 25 watts; 470 ohms ± 5%, 1/4 watt; 1K ± 5%, 1/4 watt; 470K ± 5%, 1/4 watt; 300 ohms ± 5%, 1watt; 7.1K ± 1%, 1/4 watt; 18.1K ± 1%, 1/4 watt
5.	Milliammeter	Range of 0-50ma, ±1% of full scale accuracy
6.	Digital voltmeter	Range of 0-4000 volts ± 1% of full scale accurac
7.	Oscilloscope	Rise time 100 nanoseconds or less Input impedance 10 megohms or greater Sweeptime 10 microseconds per centimeter
8.	Ammeter	Range 0-5 amps, accuracy 1% of full scale
9.	Storage capacitor	75 ± 5% microfarad at 2000 WVDC
10.	Capacitor	.047 microfarad ± 10% at 50 WVDC

4.6 Test methods and procedures.

4.6.1 Environmental.- The power supply shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the power supply shall be subjected to the tests specified in 4.6.2 through 4.6.4.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperatures specified in 3.6.

4.6.1.2 Operational temperatures.- The test shall be performed in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.7, utilizing the testing equipment indicated in 4.6.1.1. The power supply shall be exposed to each specified temperature for a minimum period of 2 hours. At the end of each 2 hour period, and while still at the specified temperature, the power supply shall be subjected to the tests in 4.6.2 through 4.6.2.4

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4.6.1.3 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926. The shock pulse shall comply with the waveform, time duration and peak "G" value specified in 3.5.

4.6.1.4 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The power supply shall be firmly mounted to the vibration table. The vibration test shall then be performed in accordance with Test Method 514 Procedure I, Part I, of MIL-STD-810, except the vibration equipment shall provide the conditions of amplitude, frequency and duration specified in 3.4.

4.6.2 Performance.- Prior to conducting the performance tests on the power supply, the following test set-up (Figure 1) shall be made and maintained during the tests, unless otherwise directed:

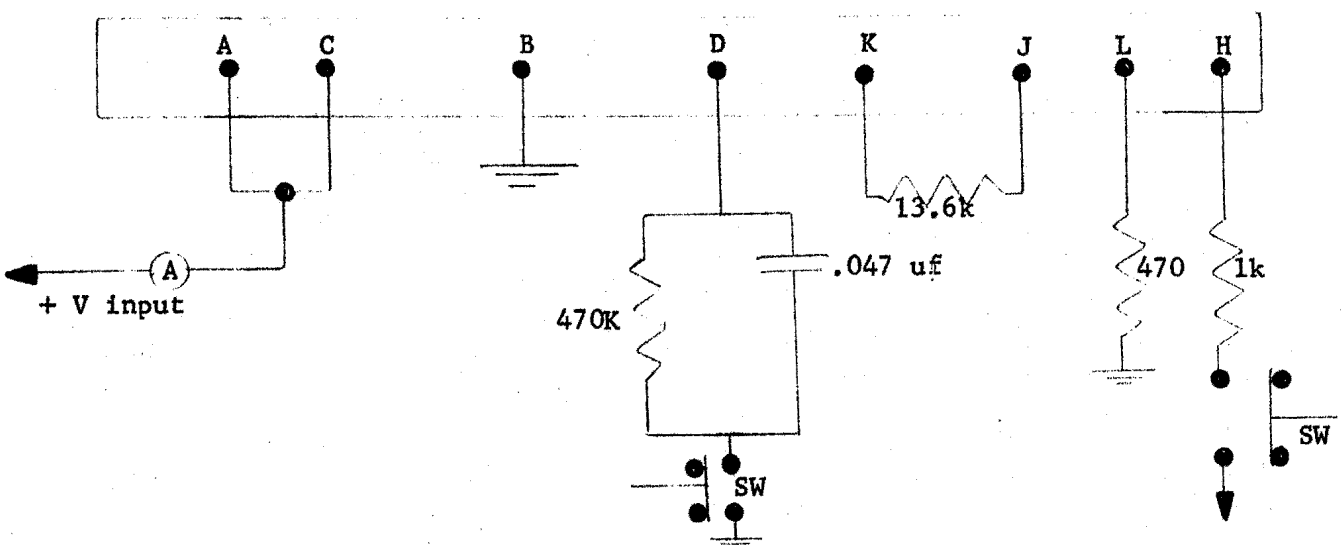


FIGURE I

a. Connect a 13.6K ± 1%, 1/4 watt resistance across contacts J and K of P-1.

b. Connect the anode of the high voltage diode to the center of contact J-1. Connect the cathode side in series with a 300 ohm, 1 watt resistance to the positive terminal of the 75 microfarad storage capacitor and connect the negative side of the storage capacitor to ground. Connect a switch for discharging the storage capacitor from the positive terminal of the capacitor to ground through a 10K ohm, 25 watt resistor.

c. Connect the positive terminal of the digital voltmeter to the positive terminal of the storage capacitor and connect the negative terminal of the digital voltmeter to ground.

d. Connect the positive side of the 0-30 volt source to Pin A of P-1.

e. Connect pin B of P-1 to ground and to the ground side of the 0-30 volt source.

f. Connect a 470 ohm, 1/4 watt resistor between pin L of P-1 and ground.

g. Connect a 470K ohm, 1/4 watt resistor in parallel with the .047 microfarad capacitor, and in series with a momentary push button switch between D of P-1 and ground.

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h. Connect a six volt d.c. source through a 1K ohm, 1/4 watt resistor, and a momentary switch to H of P-1. Connect the negative side of the 6 volt d.c. source to ground.

The power supply shall be subjected to the following tests utilizing the test equipment specified in 4.5.3. The power supply shall be operated through a minimum of 12 cycles of operation, in rapid succession, and the measured shutoff voltage recorded after each cycle of operation for the tests in 4.6.2.1 through 4.6.2.3 inclusive. Any variation between the measured values shall not exceed the tolerance specified in 3.8.1. The storage capacitor shall be completely discharged after each cycle of operation.

4.6.2.1 Capacitor charge time.- Conduct this test concurrent with the shutoff voltage test in 4.6.2.2. After applying the specified input voltage to the power supply module, momentarily ground pin D of P-1 by depressing the switch specified in the test set-up 4.6.2.g. The time interval from depression of the switch until the digital voltmeter attains its highest reading, shall not exceed the value specified in 3.8.3. A standard stop watch shall be used to measure the time interval.

4.6.2.2 Shutoff voltage,- Apply the input voltage specified in 3.8.2.a to the power supply module and conduct the tests as follows:

a. Rotate R16 potentiometer fully counterclockwise. Momentarily depress the switch specified in test set up 4.6.2.g. The shutoff voltage shall conform to the requirements of 3.8.2.a.

b. Rotate R16 potentiometer fully clockwise. Momentarily depress the switch specified in test set up 4.6.2.g. The shutoff voltage shall conform to the requirements of 3.8.2.b.

c. The measured shutoff voltage of the power supply module shall conform to the requirements of 3.8.2.a and 3.8.2.b at temperatures indicated in 3.7.

4.6.2.3 Input current.- Connect the milliammeter to pins A and C and in series with the power source, and apply an input voltage of 30 VDC. Do not depress the switch described in Test Set-Up 4.6.2.g during this test. The current draw of the power supply in the quiescent state, as indicated on the milliammeter, shall not exceed the value specified in 3.8.4.a. Momentarily depress the switch specified in Test Set Up 4.6.2.h. The current draw of the power supply, as indicated by the milliammeter shall not exceed the value specified in 3.8.4.a. Remove the milliammeter and replace it with the ammeter. Momentarily depress the switch specified in Test Set Up 4.6.2.g, and note the current draw of the power supply as indicated on the ammeter. The input current of the power supply while charging the capacitor, and as measured on the ammeter, shall not exceed the value specified in 3.8.4.b. The measured input current of the power suppl shall conform to the requirements of 3.8.4.a and 3.8.4.b at the temperatures specified in 3.7.

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4.6.2.4 Pulse output.- Connect the oscilloscope probe to pin L of P-1. Momentarily depress the switch described in Test Set-Up 4.6.2.h. The waveform observed on the oscilloscope shall have the characteristics specified in 3.8.5. The pulse output of the power supply shall conform to the requirements of 3.8.5 at the temperatures specified in 3.7.

4.6.2.5 Program shutoff voltage.- Apply the input voltage specified in 3.8.6 and conduct the test as follows:

(a) Adjust R16 potentiometer until the shutoff voltage is 1500 ± 5 VDC, with 13,600 ± 1% ohms connected between Pins J and K of P-1.

(b) Change the resistor which is connected between pins J and K of P-1 as specified in parts (a) and (b) of 3.8.6. The shutoff voltage change for each case shall conform to 3.8.6.

4.6.2.6 Final voltage setting.- Apply the input voltage specified in 3.8.7 to the power supply. Change the storage capacitor by momentarily depressing the switch described in Test Set-Up 4.6.2.g. Adjust the R16 potentiometer as required until the measured shutoff voltage of the power supply conforms to the requirements of 3.8.7.

4.6.3 Continuity.- A continuity test shall be made utilizing standard measuring equipment to determine conformance to the requirements of 3.9.

4.6.4 Interface requirements.- Perform this test using two test housing flanges. One flange designed to the maximum limit of each interface with a tolerance of 10% of total limit, and one designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limits. All pertinent interfaces shall be provided as depicted on Drawings F10540705 and F10540501. Power supply shall be capable of being properly positioned and secured to both test flanges. The power supply Connector Drawing C10540748 shall be engaged to a test connector with pertinent interfaces conforming to requirements of Drawing C10540749. All pins of power supply connector shall engage test connector receptacle to a depth of at least .09, and continuity with all test connector pins shall be maintained. The power supply jack UG1619/U shall be connected to test plug with pertinent interfaces conforming to requirements of Drawing C10541003. Continuity between jack and plug shall be indicated when fully engaged.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10540707. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The power supply is a component part of the Range Finder, Laser, An/GVS-3 and is used to charge the high voltage capacitor for energizing of the Laser cavity.

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Order of document precedence
d. Provisions for First Article testing

Custodian:

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Preparing activity:

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